Contour in Image Processing

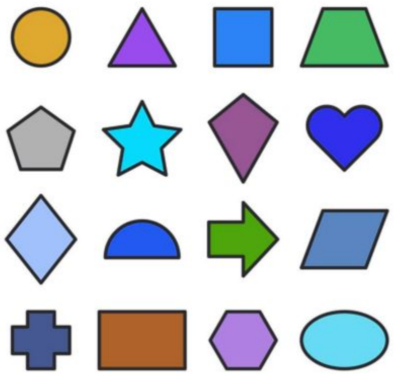
- Curves joining all continuous points with the same intensity.
- Represent object boundaries
- Difference from edges: edges show local intensity changes; contours trace shape with continuous outlines.

Extraction Steps:

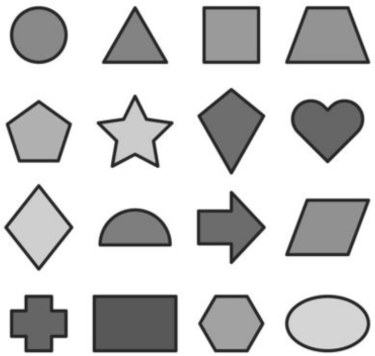
- Convert to grayscale.
- Denoise (Gaussian blur).
- Apply thresholding or Canny edge detection.
- Use cv2.findContours() in OpenCV

Example

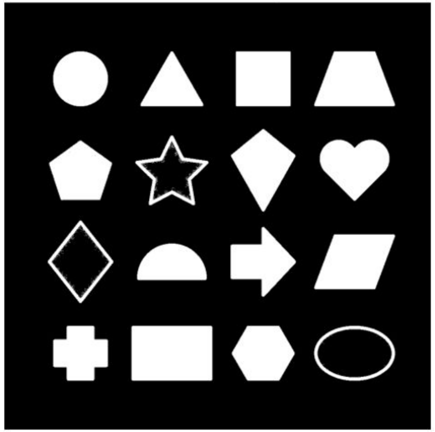
Original Image



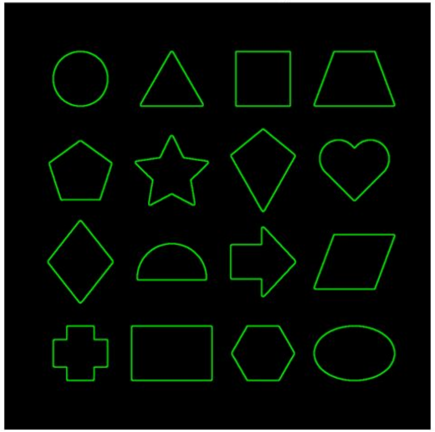
Grayscale Image



Thresholded Image



Contours on Black Background

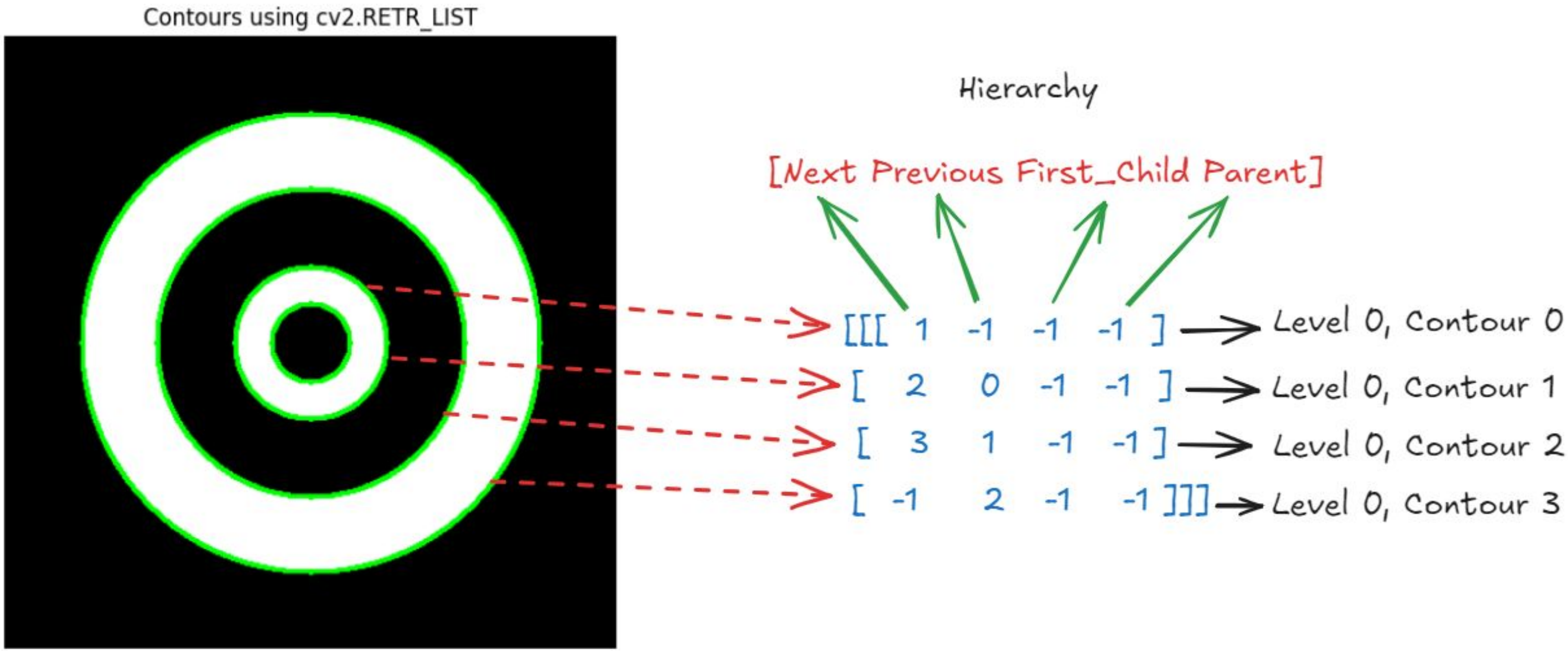


cv.findContours() function

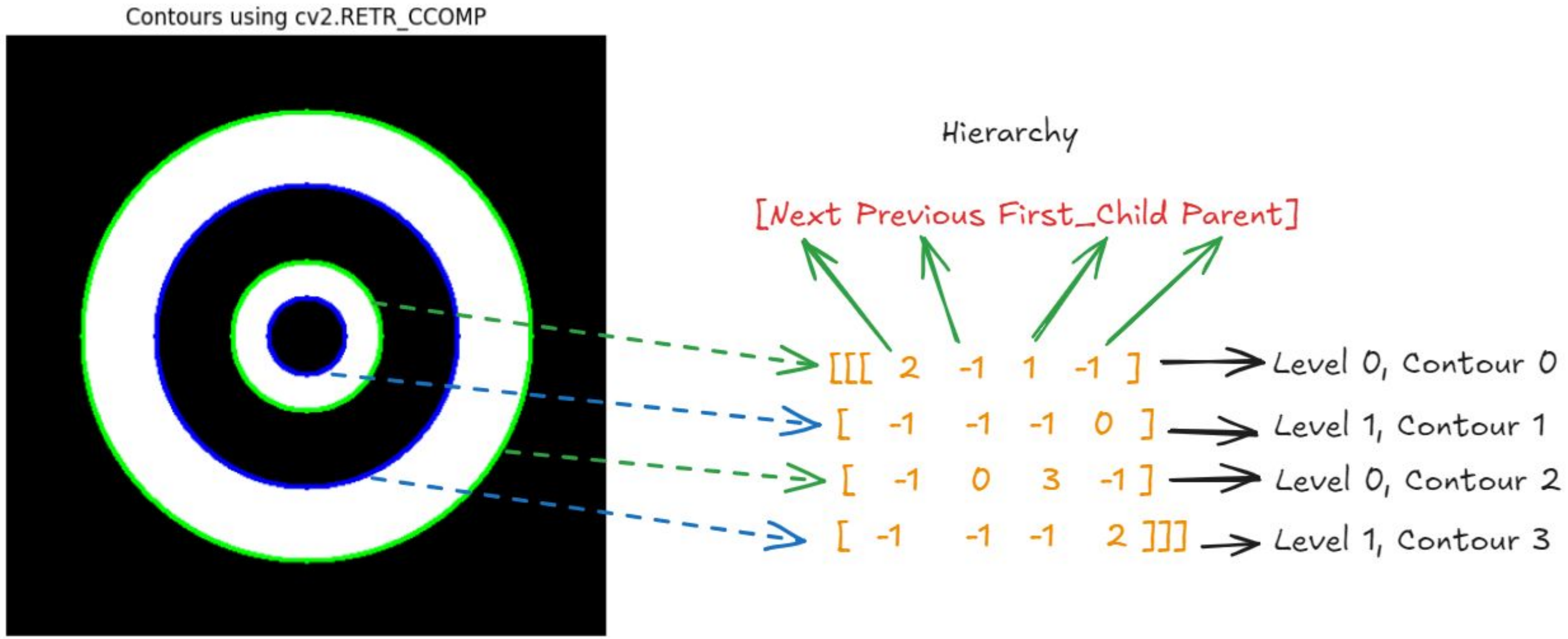
cv.findContours() function parameters,

- first one is source image,
- second is contour retrieval mode(e.g. cv2.RETR_LIST, cv2.RETR_CCOMP, RETR_TREE), RETR_LIST returns all contours in list format, RETR_CCOMP returns in two-level hierarchical format, RETR_TREE
- third is contour approximation method(e.g. cv2.CHAIN_APPROX_SIMPLE), Which it used to save memory by compressing points.

Without hierarchical format contouring



2 level hierarchical format contouring



Tree hierarchical format contouring